AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 27 · STUDENT LESSON

Predictive vs Generative

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Name the target. Compute the distribution. Check the claim.

Distinguish generated continuations, classifier outputs, calculation and retrieval without assuming plausibility establishes truth.

Keep one habit: Component contracts → conditional corpus counts → separate observation → evidence-based evaluation.

MAIN LESSON · SECTION 1

Does a plausible continuation establish a current fact?

> Separate the training target, output type and evidence used to evaluate it.

Your learning goals

I can distinguish discriminative **prediction** from generation by inspecting the requested **output**.

I can calculate an empirical next-token distribution from counted continuations.

I can separate next-token fit from factual support and downstream usefulness.

Recall prediction versus factual evidence

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- What is the difference between a model **prediction** and **evidence** that a statement is true?

Your end goal

By the end, I can distinguish a generated continuation from a **prediction** checked against the world.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Image and Voice Pipelines. Give an example and a mistake that the idea helps you avoid.

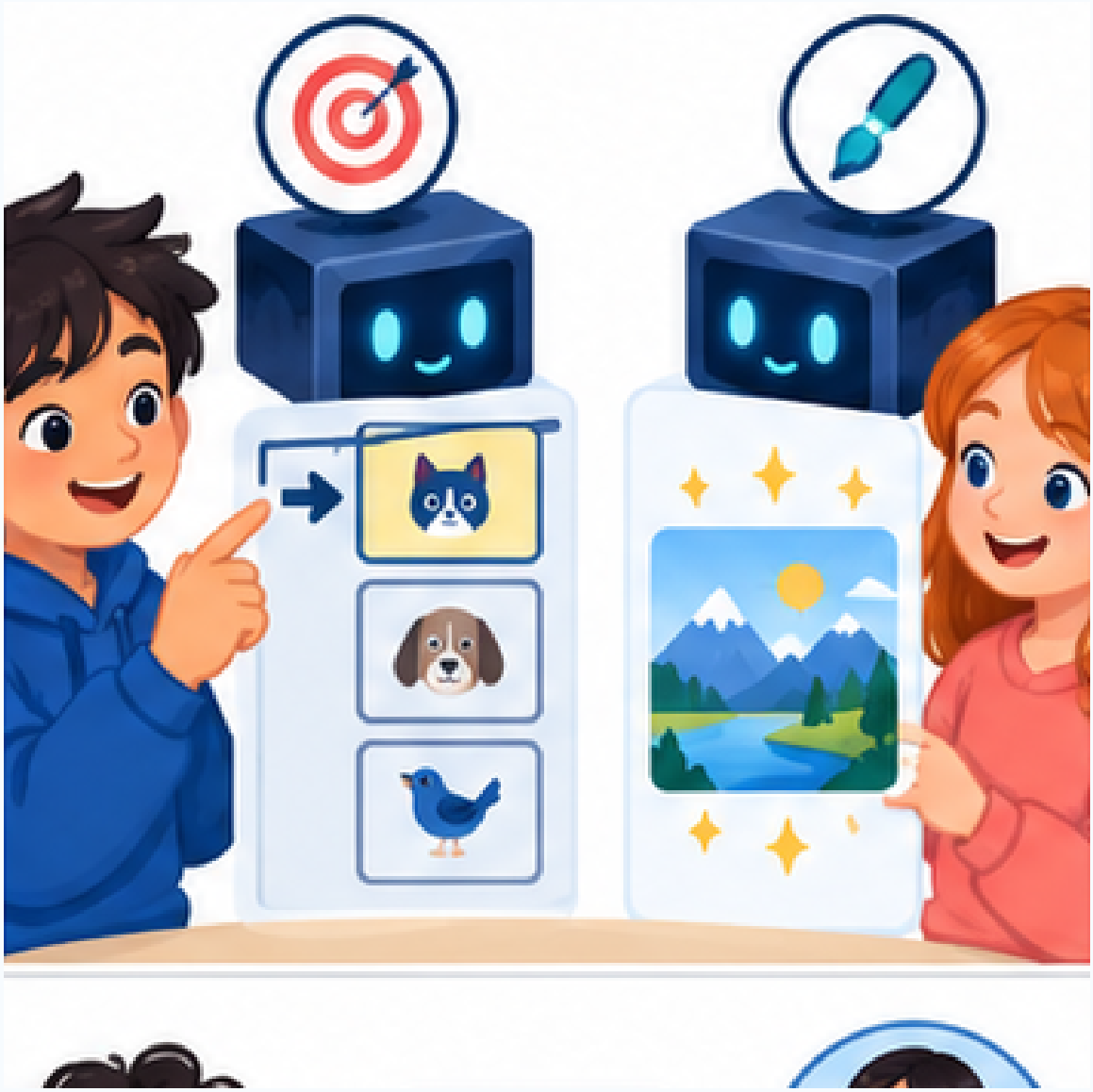
Without opening the old lesson, explain one key idea from Feedback Loops. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Similarity by Selected Traits. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 27: **Predictive** vs **Generative**

Predictive systems select a target; **generative** systems construct new **output**.
</details>



THINKER MISSION

Match problems to classifier, predictor, or generator.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week's end goal

MAIN LESSON · SECTION 2

Compare component objectives rather than product labels

> Generation makes predictions too; one application can combine several mechanisms.

Inputs, targets and permitted outputs

A classifier might map a message to the label request or report. A text generator instead produces a sequence, predicting a distribution over the next token given a prefix.

Both make **prediction**s, but the **training** targets and permitted **output**s differ. One application can combine generation with a classifier, retrieval and human approval.

Calling a whole product "**generative**" does not describe every component.

The limit of next-token fit

The objective defines what **training** attempts to fit. Next-token fit rewards matching observed continuations; it does not directly certify that an answer is true, well sourced or permitted.

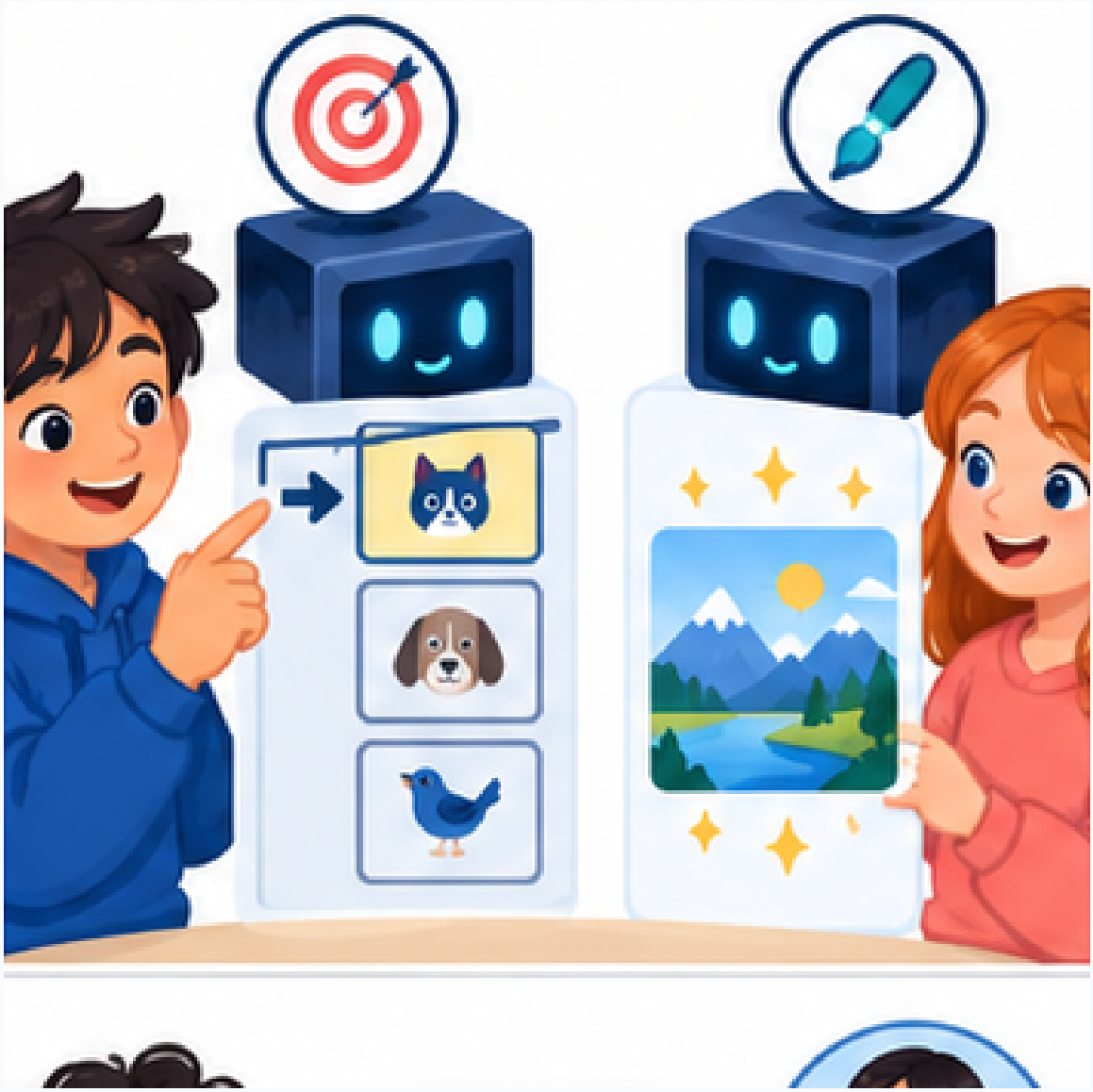
A fluent completion is still an **output** to evaluate against the task.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use **predictive** vs **generative** to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Compute a corpus distribution and label what it describes

> Counts for a prefix are not observations of the current marker.

The complete ten-continuation table

Prefix	Continuation	Count
The marker is	red	6
The marker is	blue	3
The marker is	green	1

Compute all probabilities and their sum

There are ten observed continuations. The empirical probabilities are red 6/10 = 0.6, blue 3/10 = 0.3, green 1/10 = 0.1. They are nonnegative and sum to 1.

A simple frequentist classroom model uses these relative counts as its next-token distribution. These probabilities describe the supplied corpus counts, not the current colour of any actual marker.

Trace prediction versus direct observation

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Ten corpus continuations	Count red 6, blue 3, green 1	Distribution 0.6, 0.3, 0.1
Prefix: The marker is	Select a next token	An output, not an observation
Current marker question	Check direct evidence	Colour claim needs support

The continuation model and a factual observation have different evidence sources.

Explain the evidence boundary

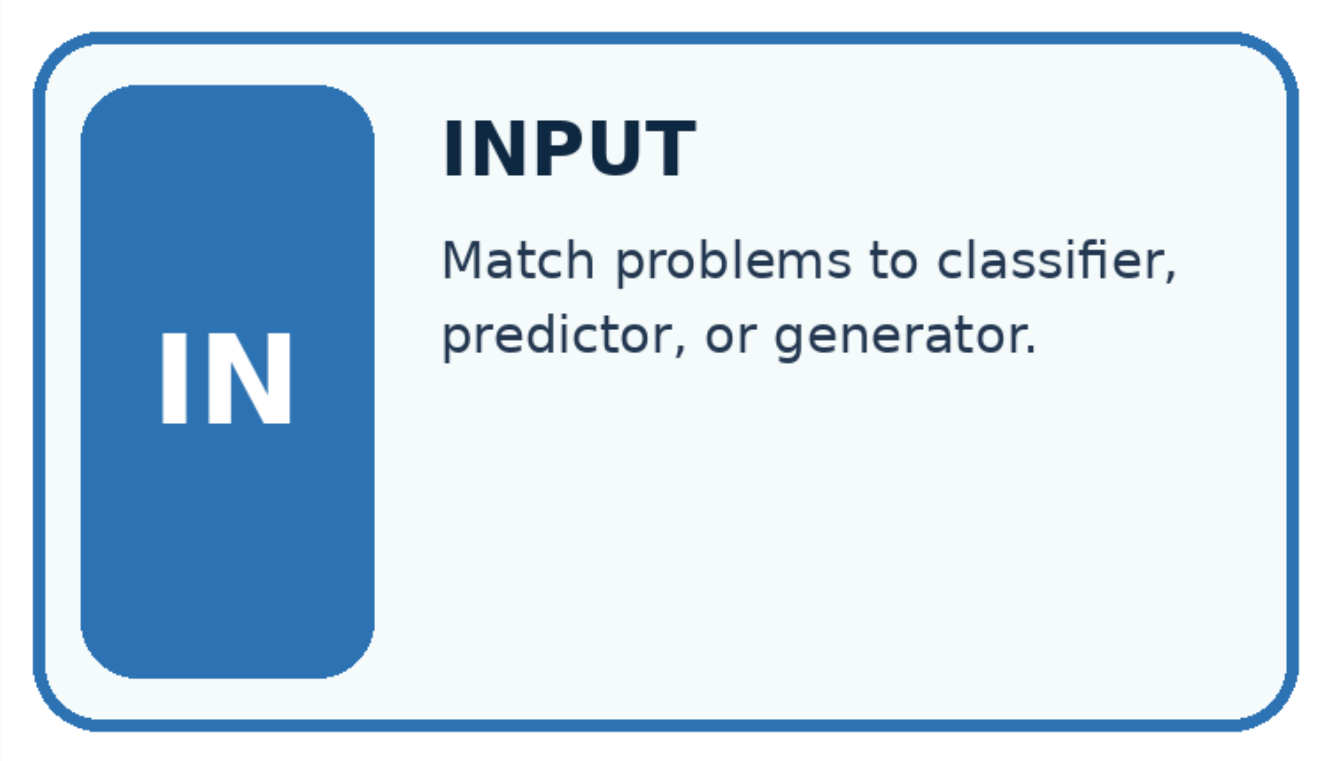
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

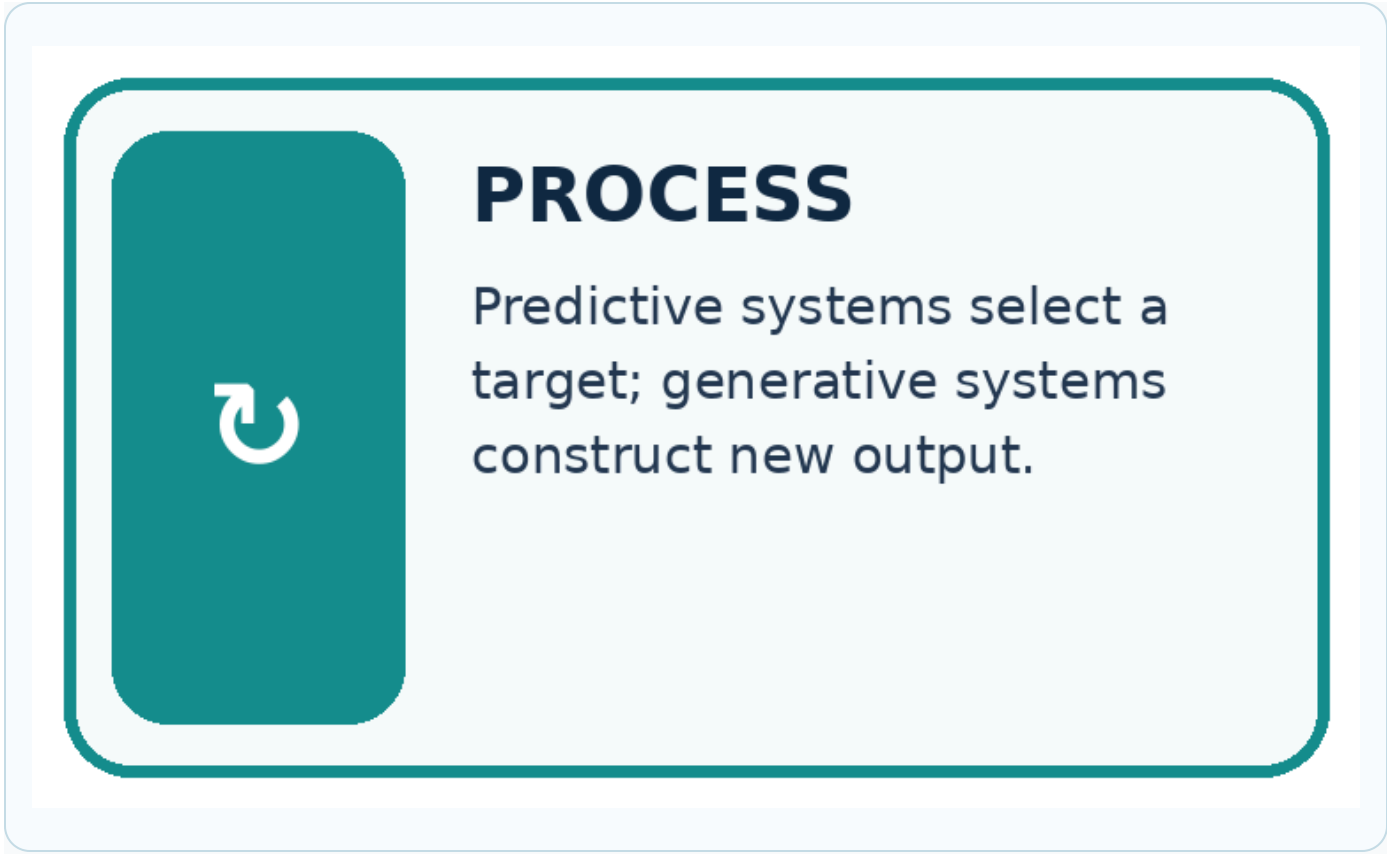
VISUAL **MECHANISM**

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓


PROCESS
Predictive systems select a target; generative systems construct new output.
OUTPUT
Match problems to classifier, predictor, or generator.
OUT

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

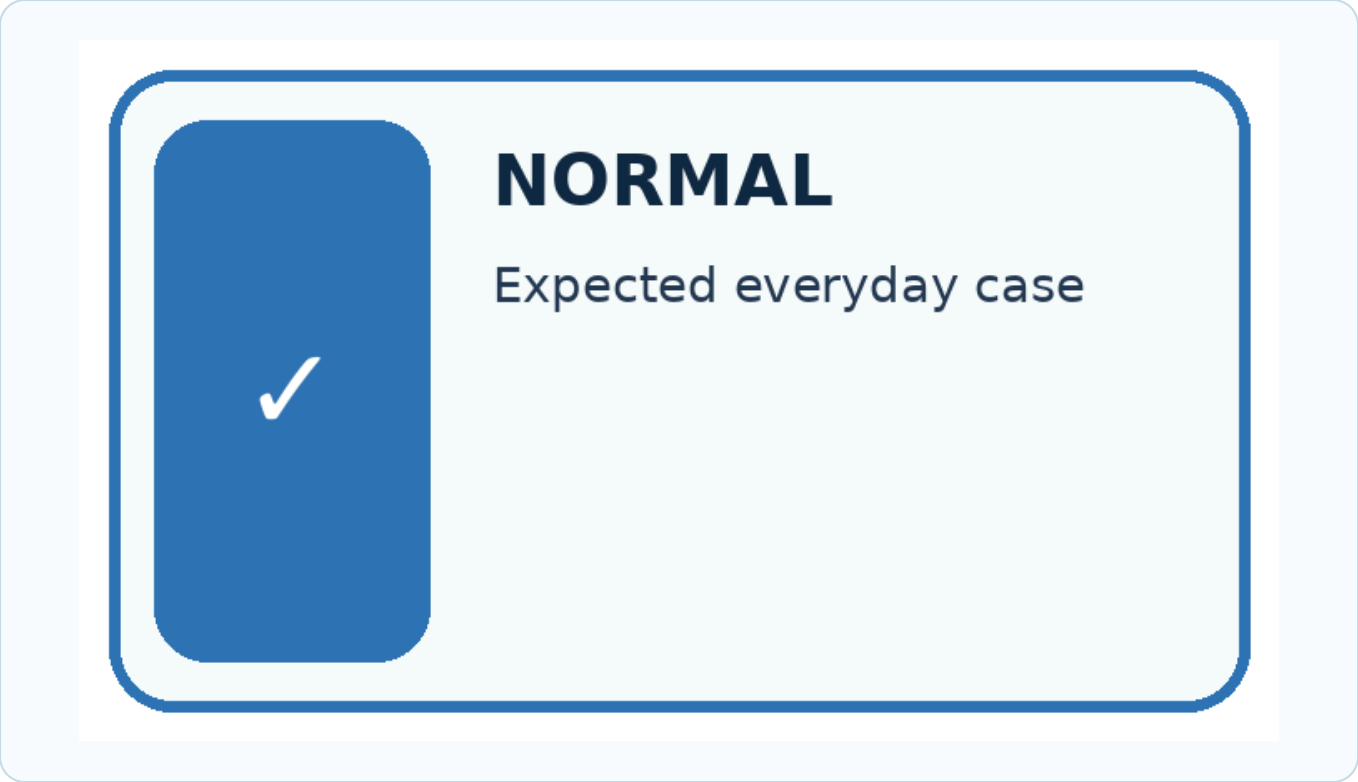
Connect the picture and the explanation

MAIN LESSON · SECTION 4

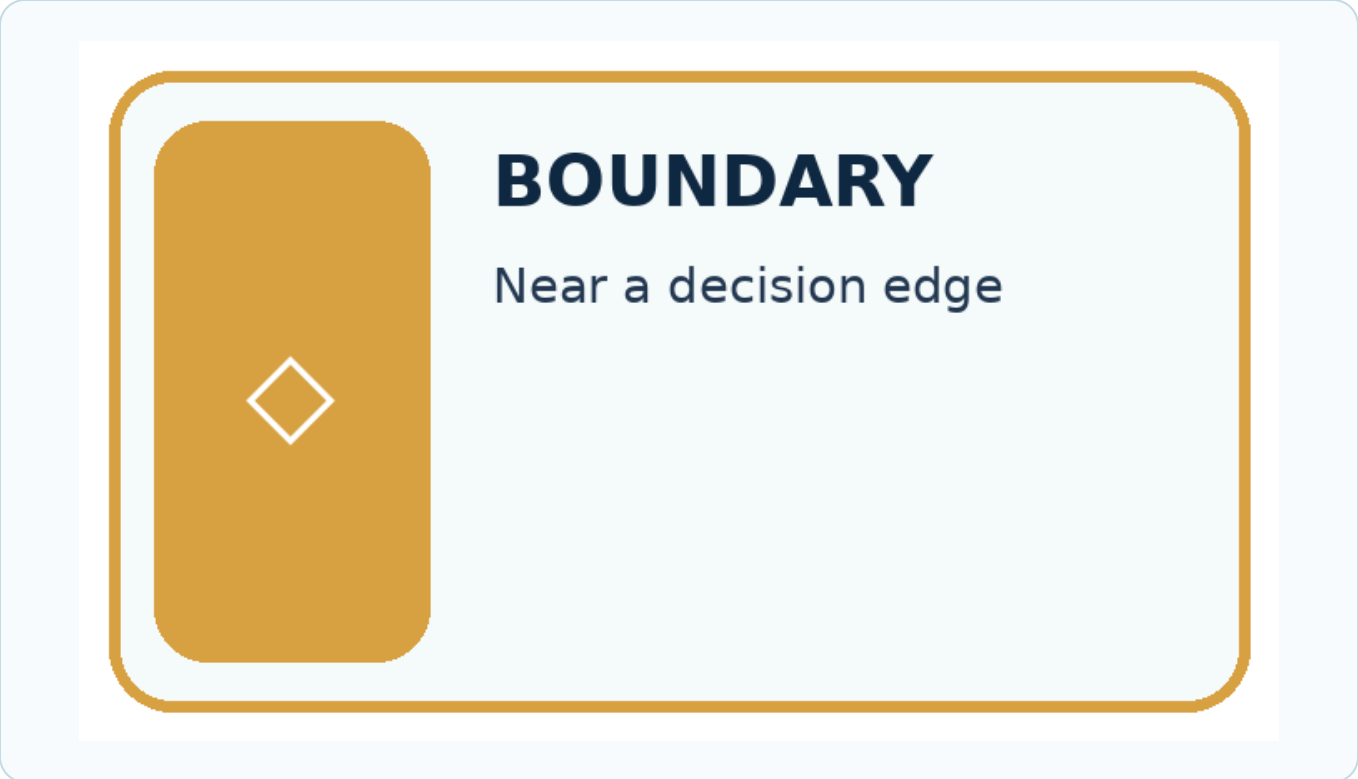
Compare model continuation and current observation

> A coincidental agreement does not turn a text generator into a sensor.

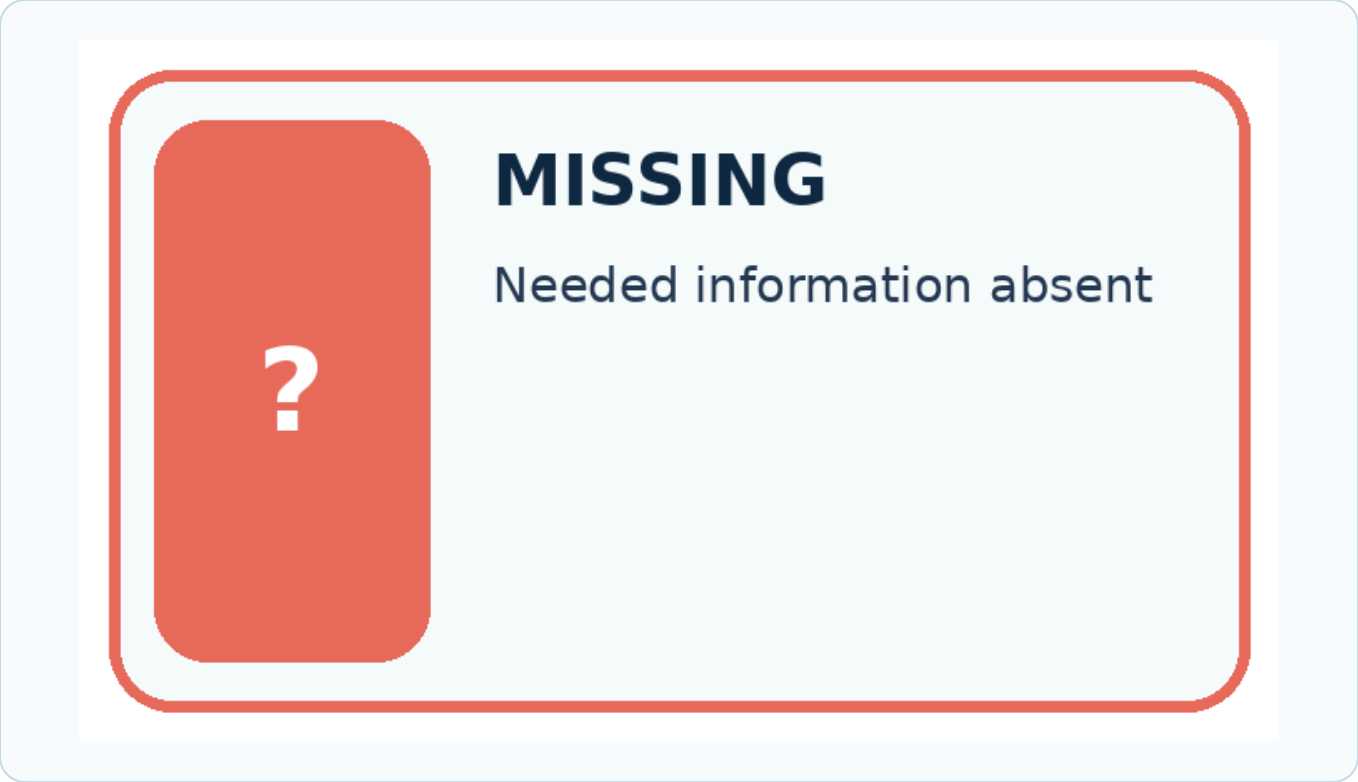
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Change the **training** table to full 2, empty 7, unknown 1.

▼ Check after predicting

Prediction: Greedy **output** becomes empty; the model assigns that token probability 0.7.

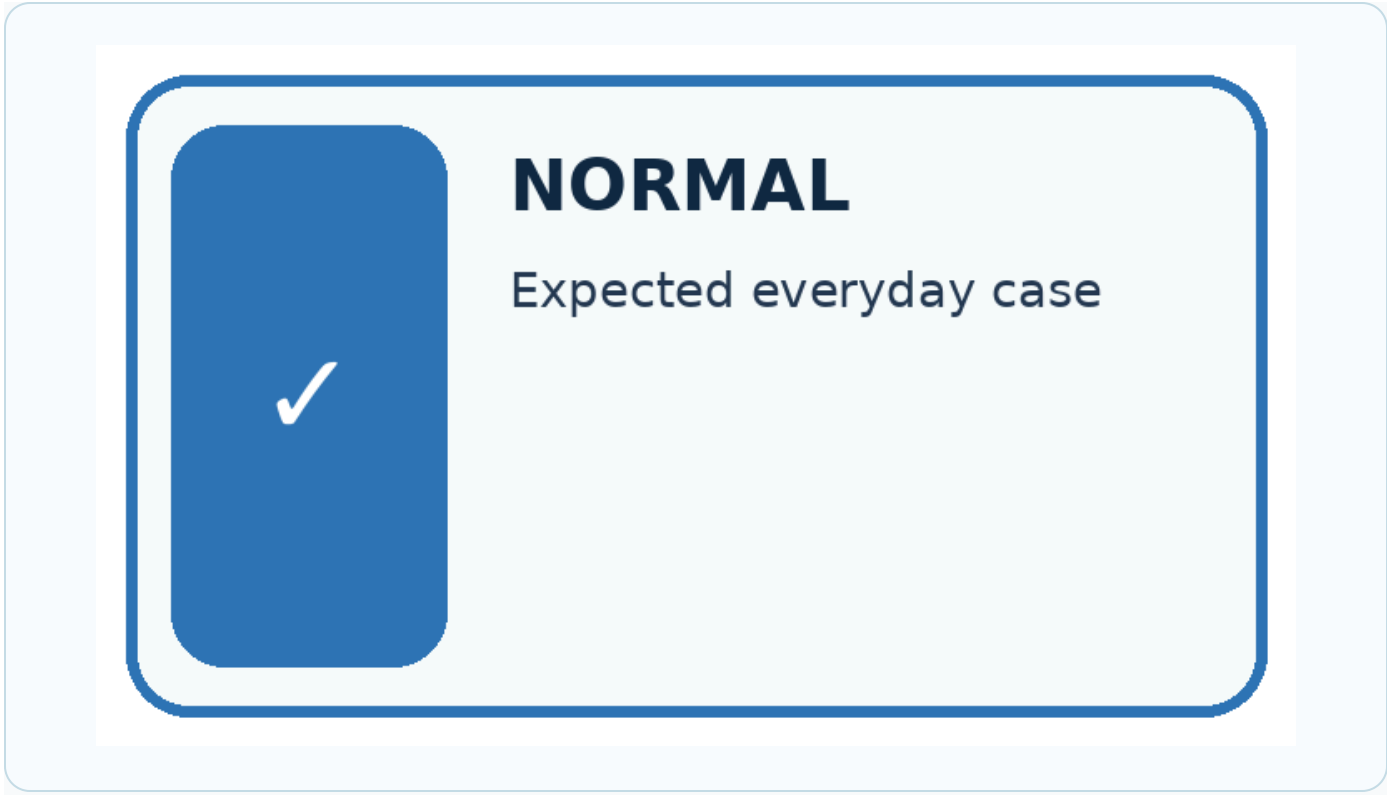
Reason: The continuation distribution changed; the independently observed box has not changed.

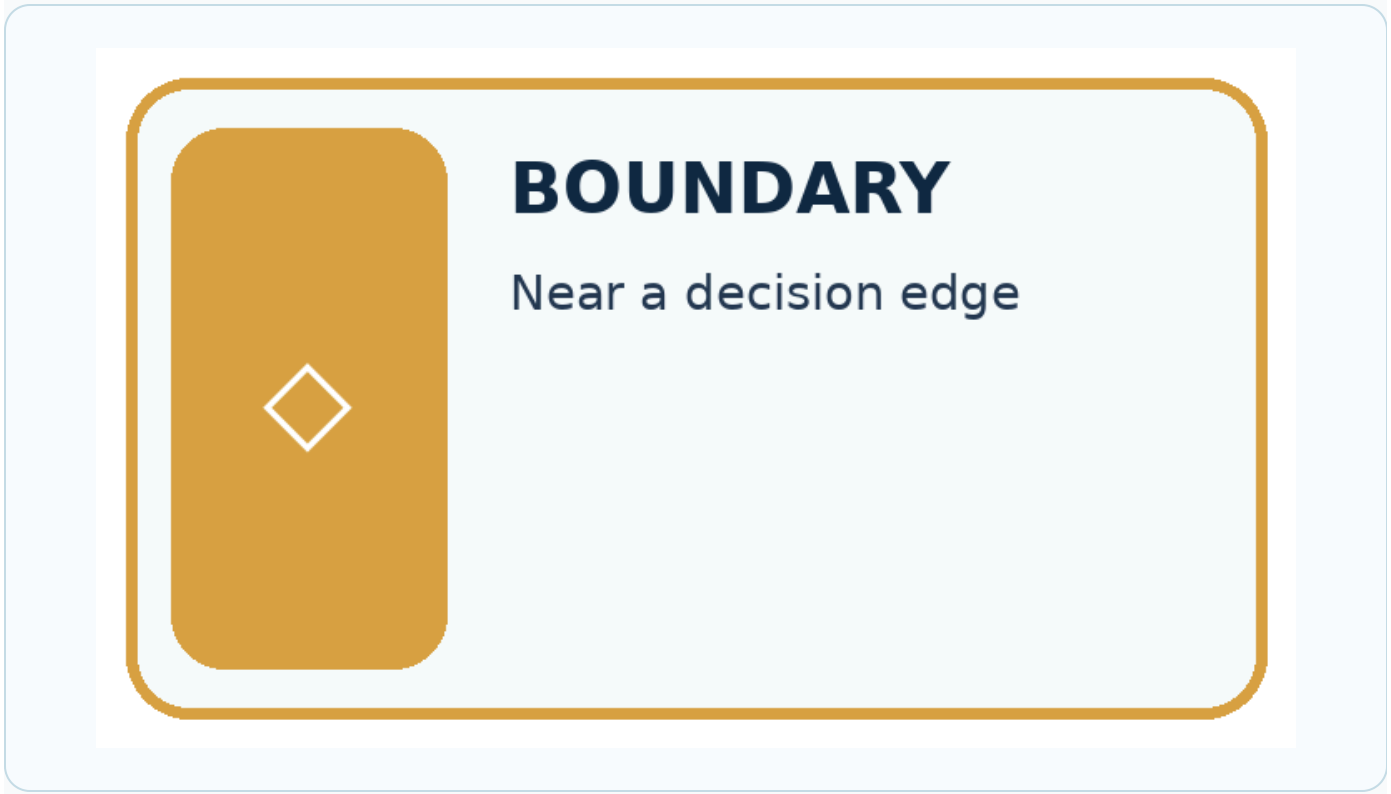
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

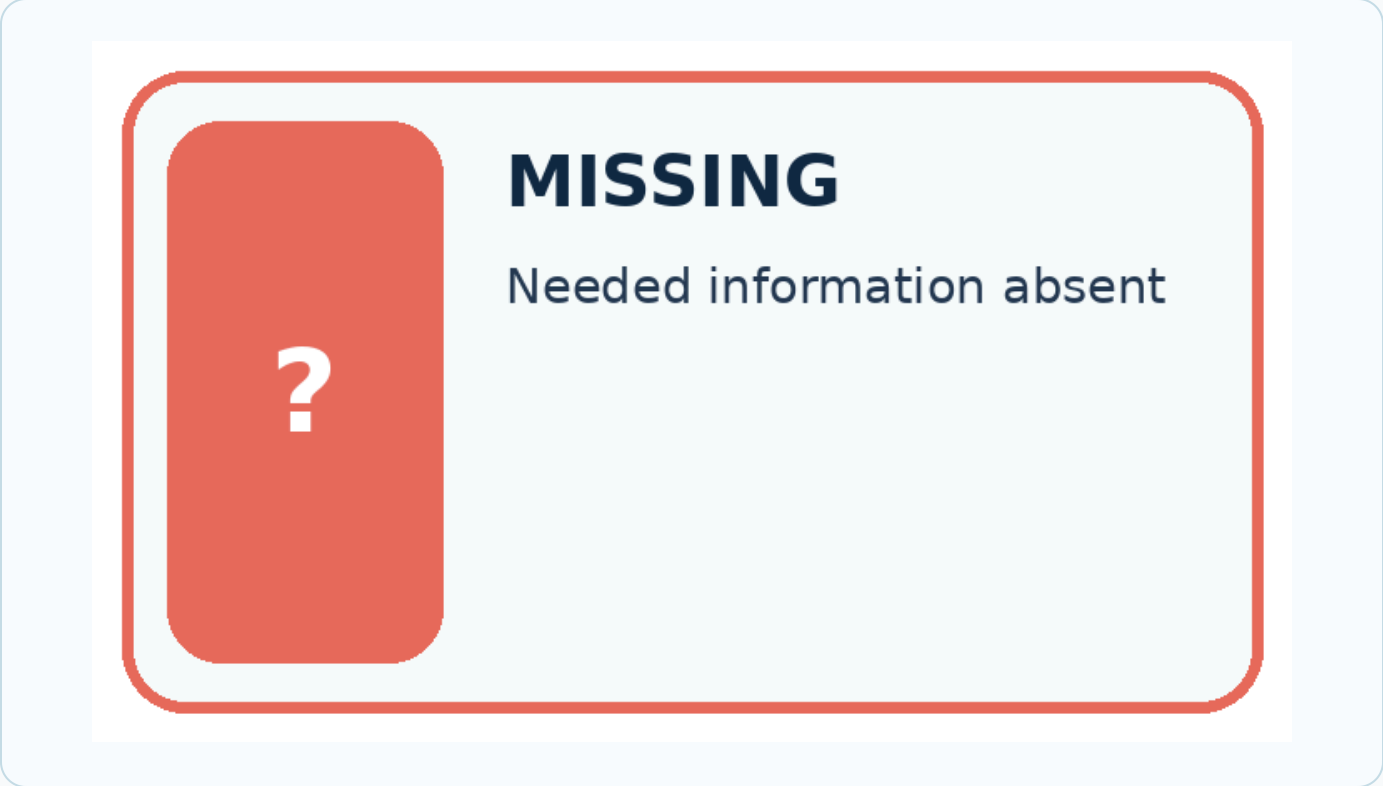
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Expand the prefix, then inspect a separate box example

> Keep the marker and box tables distinct.

Use the expanded prefix for the next step

After producing red, the next step conditions on the expanded prefix “The marker is red”. A second table might give an end token probability 0.8 and the token today probability 0.2.

A generated sequence can stop at the end token or continue. Do not reuse the first-step table unchanged after adding a token unless the model explicitly ignores the new prefix.

Separate an image-classifier target from text continuation

A separate classifier can return red as a label from an image; that is a different **input** and target from continuing text. A downstream evaluation asks whether the system correctly reports the current marker.

It must include ground truth or sources and appropriate failure handling, not just count plausible phrases.

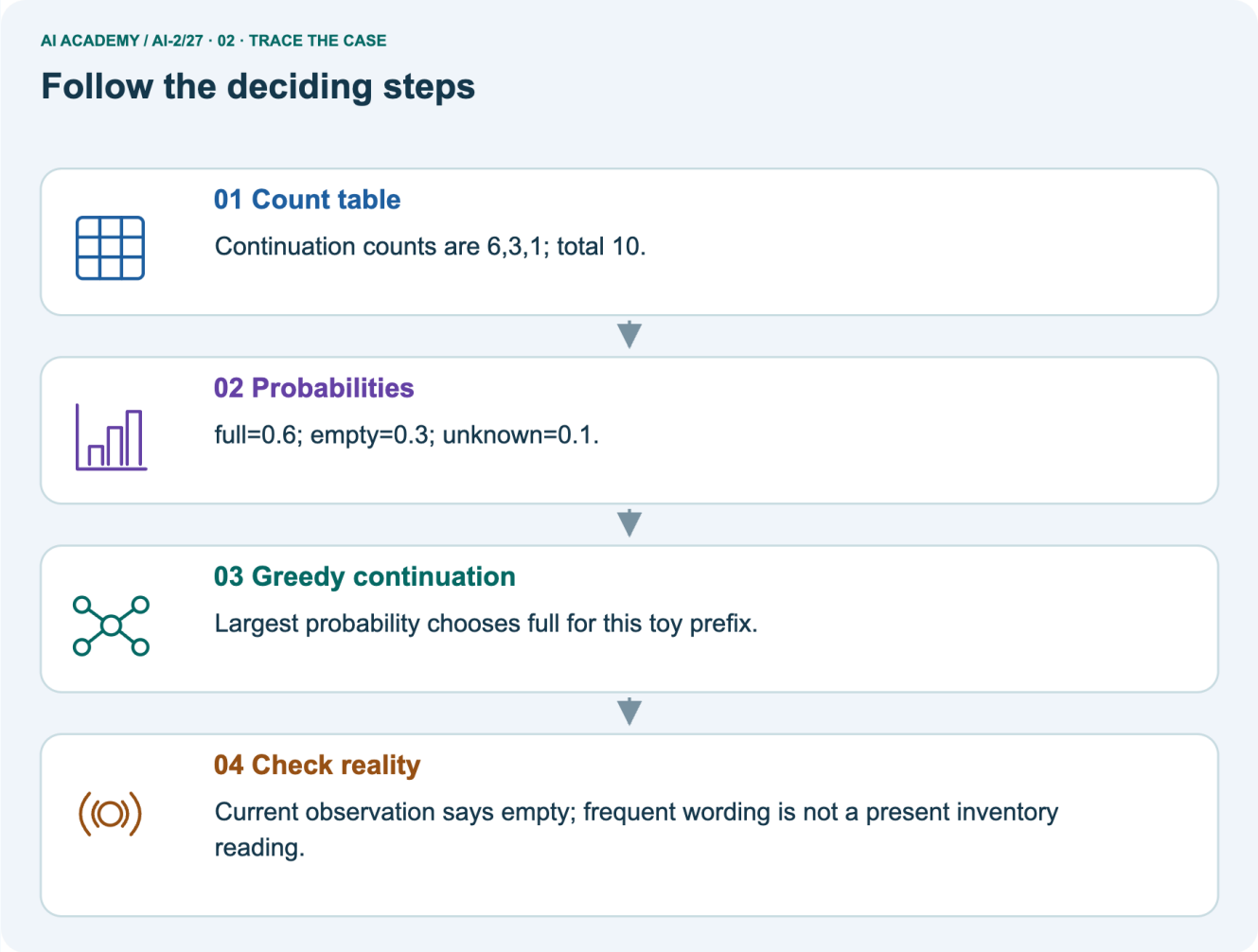
Read the separate box corpus and current observation

A fictional toy generator predicts the next word after “The box is”.

Observed **training** continuations: full 6, empty 3, unknown 1.

Use counts divided by total 10. The current physical box is separately checked and found empty.

Trace relative counts, greedy output and the factual conflict



Follow the labelled picture with the matching explanation. Open the vector diagram

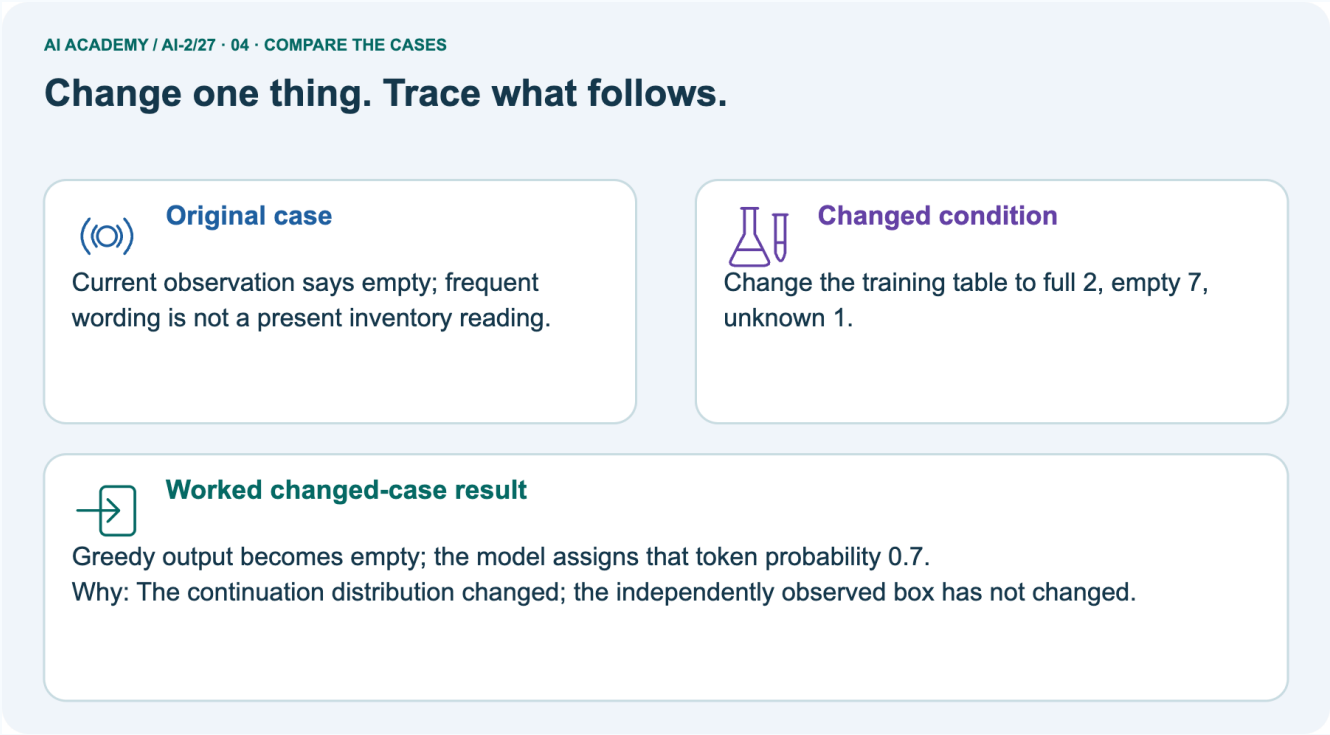
The table predicts a continuation in a text exercise, not the contents of the current box. Generating the most frequent token is a model-defined decision. A factual inventory statement needs current **evidence**.

The numerical table is intentionally small; real **generative** systems condition on richer context and do not simply consult these three counts.

Predict the changed corpus without changing the physical box

- Change one thing: Change the **training** table to full 2, empty 7, unknown 1. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain coincidence and the model-scope limit

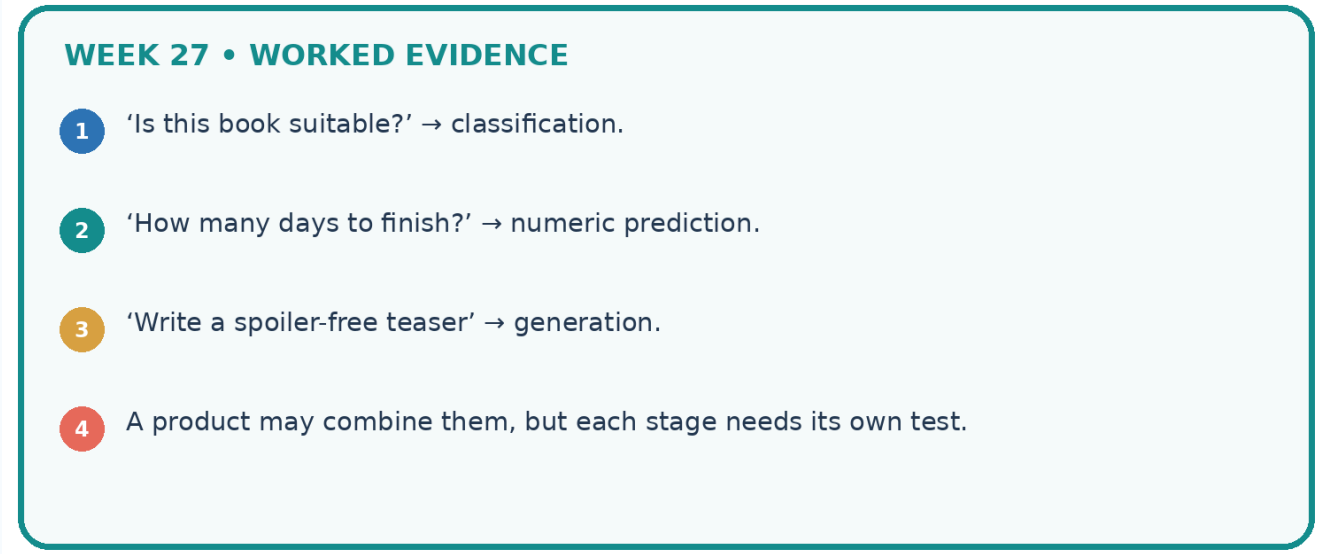
- Read the labelled picture: Does matching the physical box in the changed table prove the generator measures boxes?
- A one-step toy distribution does not describe every **predictive** or **generative** model or validate factual accuracy.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. 'Is this book suitable?' → classification.

2. 'How many days to finish?' → numeric **prediction**.

3. 'Write a spoiler-free teaser' → generation.

4. A product may combine them, but each stage needs its own test.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

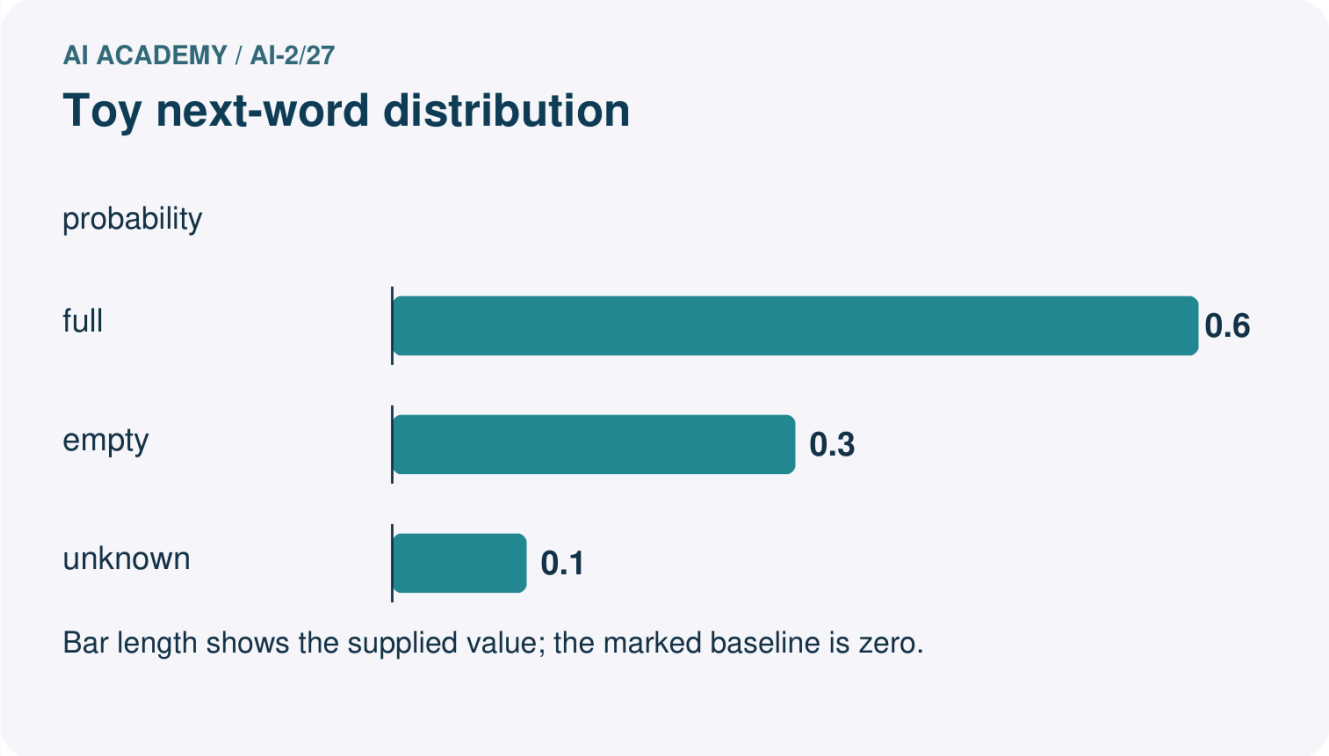
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Predictive vs Generative

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/27

Predictive vs Generative

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Count table

Continuation counts are 6,3,1; total 10.

2. Probabilities

full=0.6; empty=0.3; unknown=0.1.

3. Greedy continuation

Largest probability chooses full for this toy prefix.

4. Check reality

Current observation says empty; frequent wording is not a present inventory reading.

CHANGE ONE THING

Change the training table to full 2, empty 7, unknown 1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Count table	Continuation counts are 6,3,1; total 10.
2. Probabilities	full=0.6; empty=0.3; unknown=0.1.
3. Greedy continuation	Largest probability chooses full for this toy prefix.
4. Check reality	Current observation says empty; frequent wording is not a present inventory reading.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Evaluate fit, task compliance and factual support separately

> A fluent statement is still a claim to check against the appropriate evidence.

State the three different evaluation questions

Keep three questions separate: did the model fit continuation data; did the **output** satisfy the requested format or task; and are factual claims supported?

A system that drafts a fictional description has different criteria from one reporting a live measurement. Record the source and time when a claim refers to changing conditions.

No AI account is needed to reason about these distinctions.

Challenge highest probability proves current truth

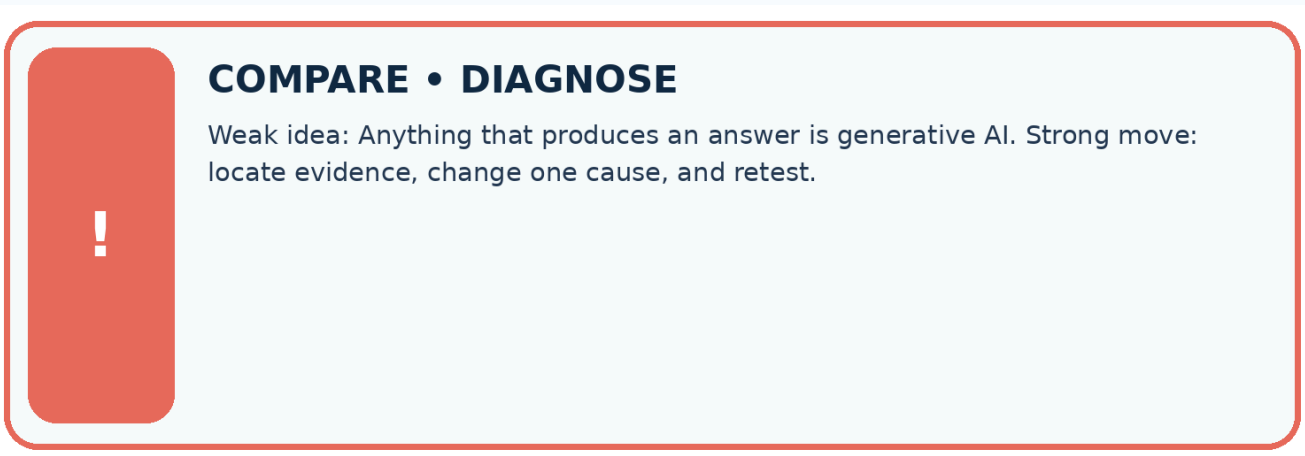
> **Exit: a team says “the highest-probability continuation proves the current fact”. Name the evidence they actually have and the evidence still needed.**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.



Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 8

Use the supplied door corpus as a controlled practice case

> Do not substitute the next-prefix probabilities for the original alternatives.

Read the separate door counts and sensor observation

Supported case: the prefix “The door is” is followed in a tiny corpus by open four times and closed once. The empirical probabilities are 4/5 and 1/5. A current sensor reports closed.

Selecting the most frequent continuation open would be plausible under the corpus model but inconsistent with the sensor **observation**.

Try both probabilities and the factual limit

Guided: compute both probabilities and explain why selecting open does not establish the current door state.

Use the marker table for W2

Use workbook W2 for the connected example "**Generative** and **predictive** objectives". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Repair generative models do not predict

- Use the worked case for **Generative** and **predictive** objectives. Proposed explanation: "**Generative** models do not predict". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the changed box-table coincidence

Does matching the physical box in the changed table prove the generator measures boxes?

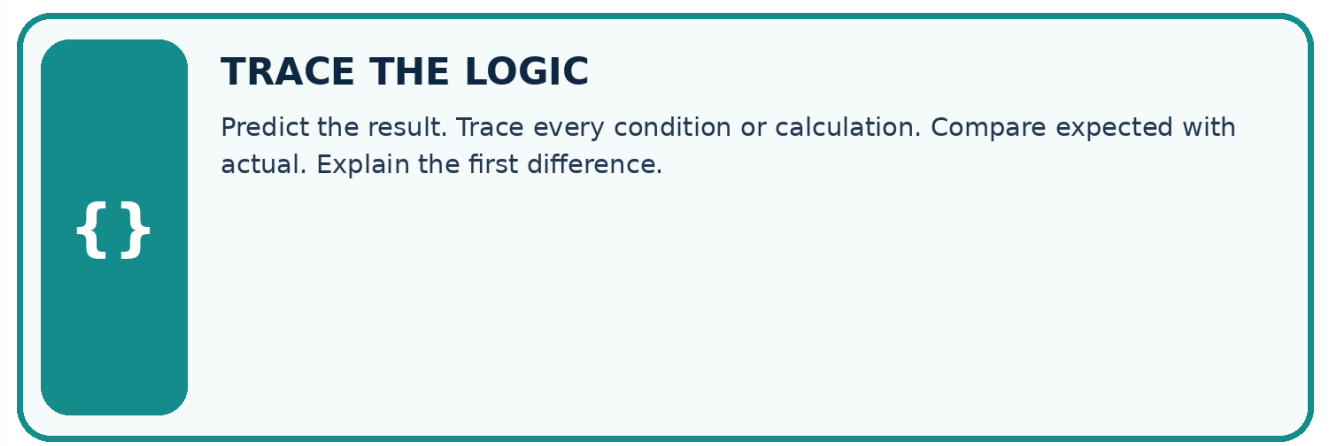
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Practise with less help: Compute the distribution

Check your reasoning

Explain the picture

> **My explanation using a pictured step and the given facts**

MAIN LESSON · SECTION 9

Audit nine complete component cards

> Name the evaluation evidence rather than inventing performance.

The nine supplied task contracts

Card	Supplied component and output	Evaluation evidence to request
1	Arithmetic procedure computes24/6	Correct calculation under the stated inputs
2	Catalogue lookup returns the record for a given ID	Correct record identity and version
3	Learned classifier maps description to genre label	Separate examples with reviewed genre references
4	Learned regression predicts numeric pages	Separate reference counts, errors and units

Card	Supplied component and output	Evaluation evidence to request
5	Toy prefix model uses the supplied marker count table	Declared distribution/count fit, not present marker truth
6	Generator writes a new blurb from supplied book facts	Task format and support for each factual claim
7	Sequence-model transcriber emits words from supplied audio	Available reference audio/field transcript and stage errors
8	Declared rule ranks eligible activities by weighted score	Eligibility, score components and tie policy
9	App classifies a request, retrieves a record, then writes an answer	Each stage, source identity and final claim support

Sort by target, explain failures and design checks

1. For each of the nine supplied cards name input, permitted output and the component's stated mechanism. Distinguish deterministic calculation/retrieval/ranking from learned label or numeric prediction and sequence generation; do not assume a whole app has one category. Some sequence tasks constrain the output closely. Propose a concrete failure and a check for each using the declared evidence; the table supplies task descriptions, not real model results. Calculate the marker empirical distribution from its own table, then separately interpret current observations. No AI service, training run or actual sensor measurement is performed. Reserve the signal corpus for independent W4.

Keep the evidence record and limits

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

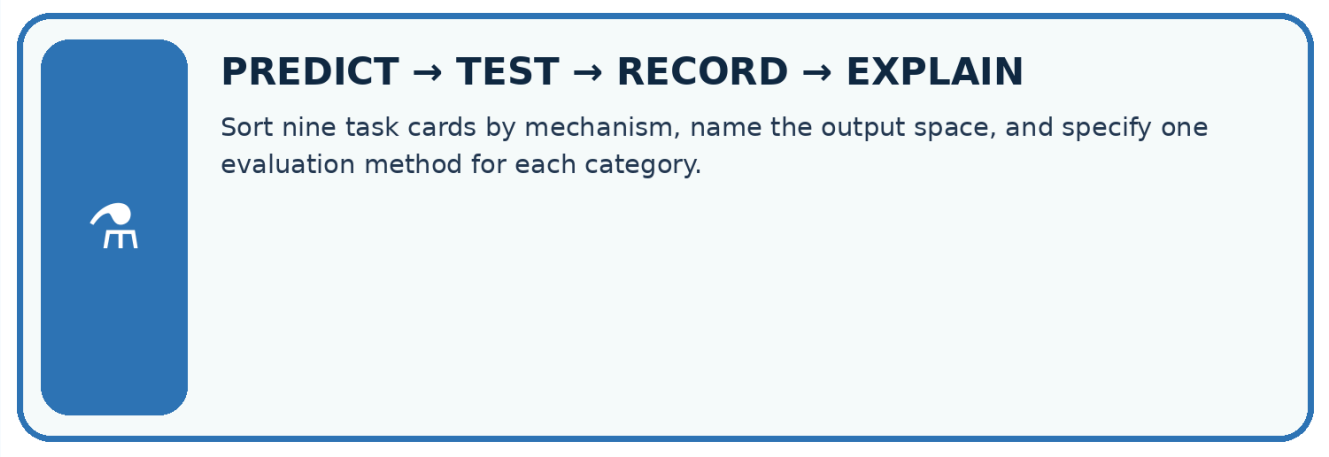
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the supplied nine component cards. State input/**output** contracts, distinguish the stated **mechanism**s and propose evidence-based checks. No real model **output**s or performance scores are supplied for these task cards.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: analyse the fresh signal corpus and observation

> Retain the difference between most frequent continuation and current measurement.

The independent counts and current evidence

For prefix The signal is, counts are high 2, low 5, unknown 3. Compute all probabilities and the most frequent continuation. A current measurement reports high.

State what the most frequent continuation does and does not establish.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after the attempt

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Analyse a new corpus independently

Week 27 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

One model labels a review positive or negative; another writes a new review. Which classifies and which generates? Must generated text be true?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Keep scoring **predictive** and explanations templated or carefully generated and verified.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A tool retrieves a book description, predicts a genre, then writes a new blurb. Identify each operation and give one distinct failure to check at each stage.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Explain each recommender component by its task

> Generation does not validate retrieval, scoring or a factual explanation.

This week’s concrete contribution

For the fictional recommender identify separately the eligibility/score procedure, any learned category predictor, retrieved catalogue evidence and templated or generated explanation. A scoring rule does not become learned merely because another component generates prose. Require every factual blurb claim to agree with approved book facts and report unresolved information rather than fabricating it.

Use approved book facts for recommender explanations; generated fluency is not a replacement for catalogue **evidence**.

Keep the checkpoint and human choice

THIS WEEK’S CHECKPOINT Keep scoring **predictive** and explanations templated or carefully generated and verified.

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

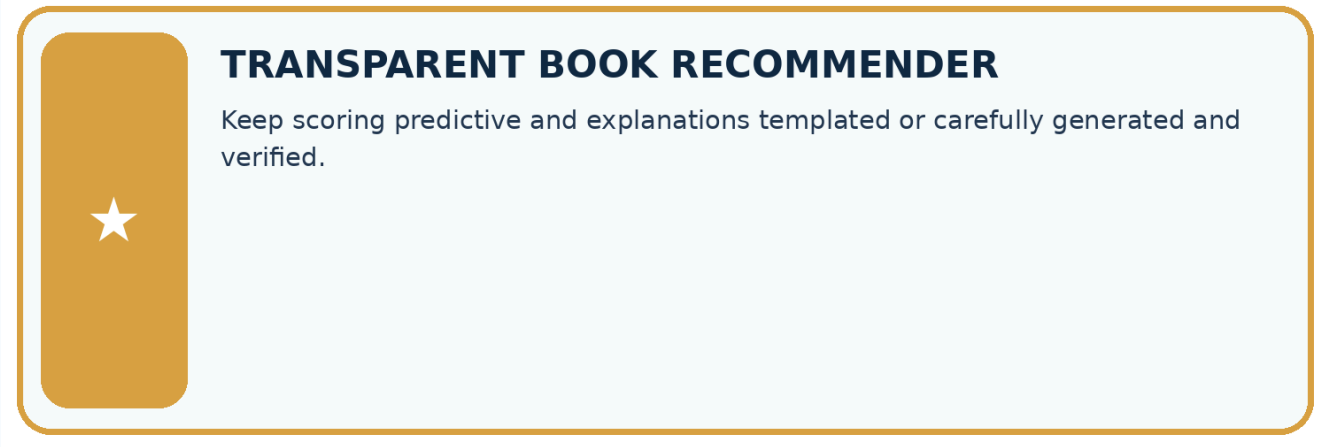
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: name the objective, output and evidence for the claim

> Do not replace factual evaluation with plausibility.

Explain without notes

- Close your notes. Explain this goal in your own words: Distinguish discriminative **prediction** from generation by inspecting the requested **output**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
predictive	Estimating a target from supplied inputs.	
generative	Producing new content from learned patterns and supplied conditions.	
output space	The set or range of outputs a system can produce for a task.	
fit for purpose	Suitable for the stated users, task, conditions and consequences.	

Terms for the connected explanation

Word	Meaning
Generative objective	A training aim concerned with producing or modelling possible outputs
Discriminative objective	A training aim mapping inputs to a specified target such as a label
Conditional distribution	Probabilities for alternatives given stated information

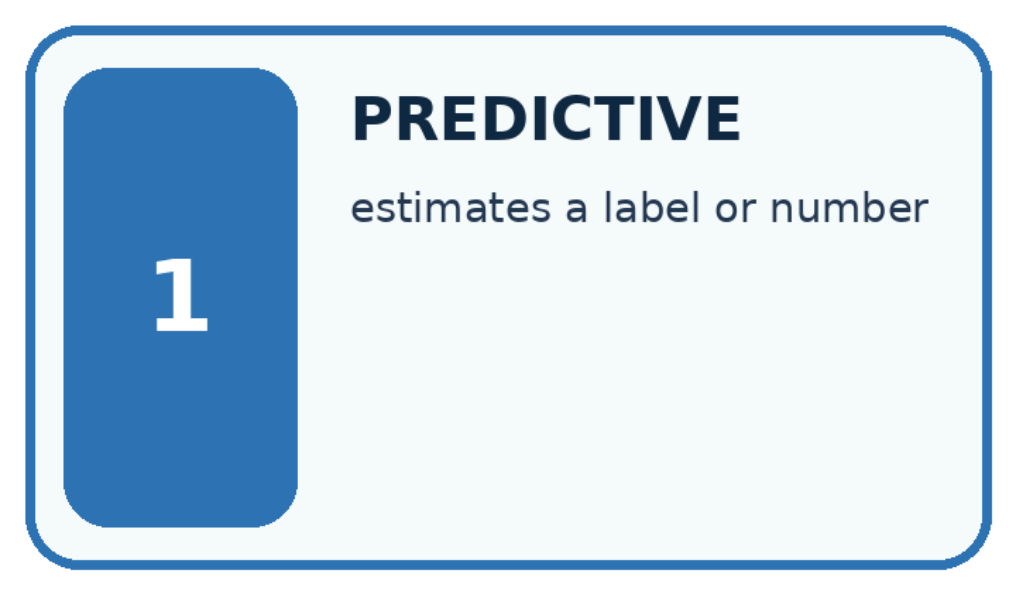
Word	Meaning
Autoregressive generation	Producing a sequence one next item at a time using preceding items

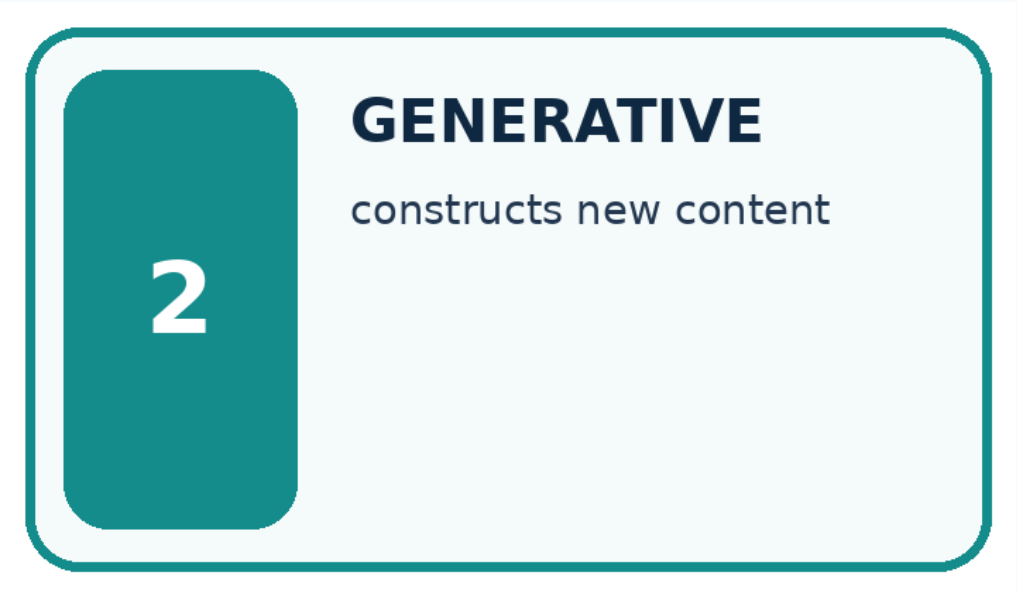
▼ Earlier wording, alternatives and picture transcripts

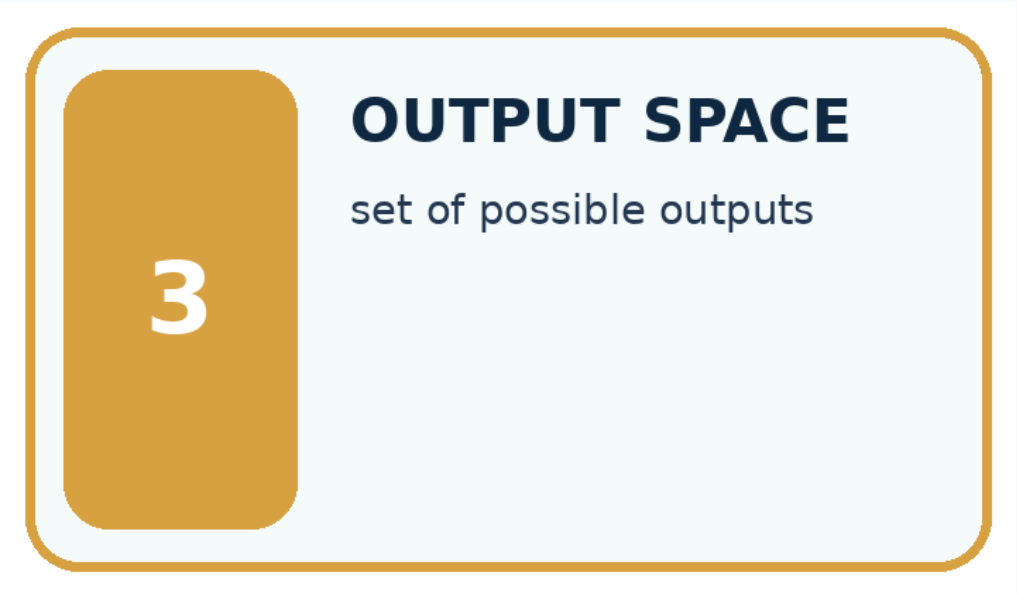
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
GENERATIVE
constructs new content


3
OUTPUT SPACE
set of possible outputs



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

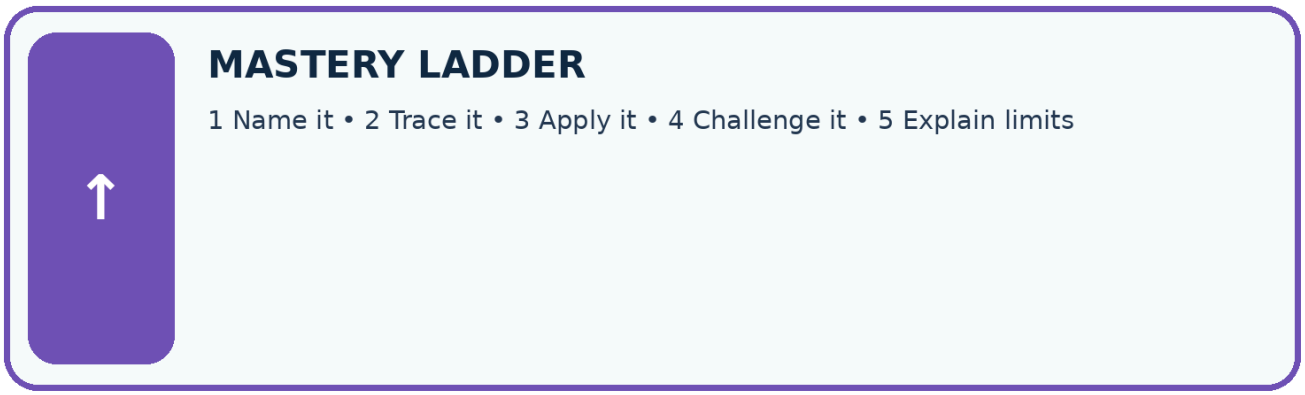
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Analyse a new corpus independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

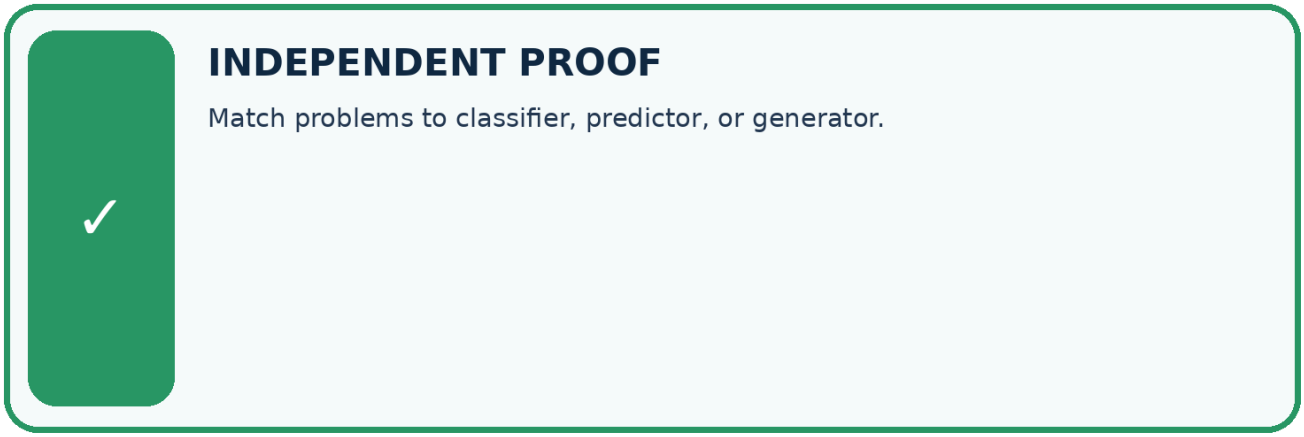
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Analyse a new corpus independently is the selected independent case.



EXPLANATION CHECK State the component objective and conditioning prefix, compute probabilities from the correct table and evaluate factual claims against supplied observations.

Write or draw your reasoning and evidence.

▼ **Extra practice: Three operations in one book helper**

REFERENCE · SECTION 13

Extra practice: Three operations in one book helper

> Try the question before opening its worked explanation.

Read the supplied case and question

A labels genre; B writes a new blurb; C computes 24/6.

Distinguish operations.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

A predicts a category, B generates text, C calculates 4. Products may combine them.

Evidence or stage	Value or result
Classify	Select a genre label
Generate	Write a new blurb
Calculate	24 divided by 6 equals 4

▼ Earlier wording, alternatives and picture transcripts

Week 27 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Three operations in one book helper

A labels genre; B writes a new blurb; C computes 24/6.

Distinguish operations.

A predicts a category, B generates text, C calculates 4. Products may combine them.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Distinguish discriminative prediction from generation by inspecting the requested output	Explain the deciding step, show a trace or test, and state one limit.
Calculate an empirical next-token distribution from counted continuations	Explain the deciding step, show a trace or test, and state one limit.
Separate next-token fit from factual support and downstream usefulness	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: A fictional toy generator predicts the next word after “The box is”.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional toy generator predicts the next word after “The box is”.
- Which labelled step matches this detail? Current **observation** says empty; frequent wording is not a present inventory reading.
- 1. Probabilities
- 2. Count table
- 3. Greedy continuation
- 4. Check reality

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Check reality Current **observation** says empty; frequent wording is not a present inventory reading.

The table predicts a continuation in a text exercise, not the contents of the current box. Generating the most frequent token is a model-defined decision. A factual inventory statement needs current **evidence**.

The numerical table is intentionally small; real **generative** systems condition on richer context and do not simply consult these three counts.

Changed-case question

- Changed-case practice: Change the **training** table to full 2, empty 7, unknown 1. Which conclusion is supported by the supplied facts?
- 1. Greedy **output** becomes empty; the model assigns that token probability 0.7.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Greedy **output** becomes empty; the model assigns that token probability 0.7. The continuation distribution changed; the independently observed box has not changed.

Repair and save your work

Review Image and Voice Pipelines (ai-2/26). Goal: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**.

Short practice: Which stage first fails, and why should the faulty transcript not be silently overwritten?. Return to this week and explain what you changed.

Review Feedback Loops (ai-2/24). Goal: Implement a deterministic recommender. Short practice: What extra field would reveal why comparing raw clicks is misleading?. Return to this week and explain what you changed.

Review Similarity by Selected Traits (ai-2/21). Goal: Represent items with numeric features. Short practice: Explain why B ceases to be a closest option without any candidate data changing..

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

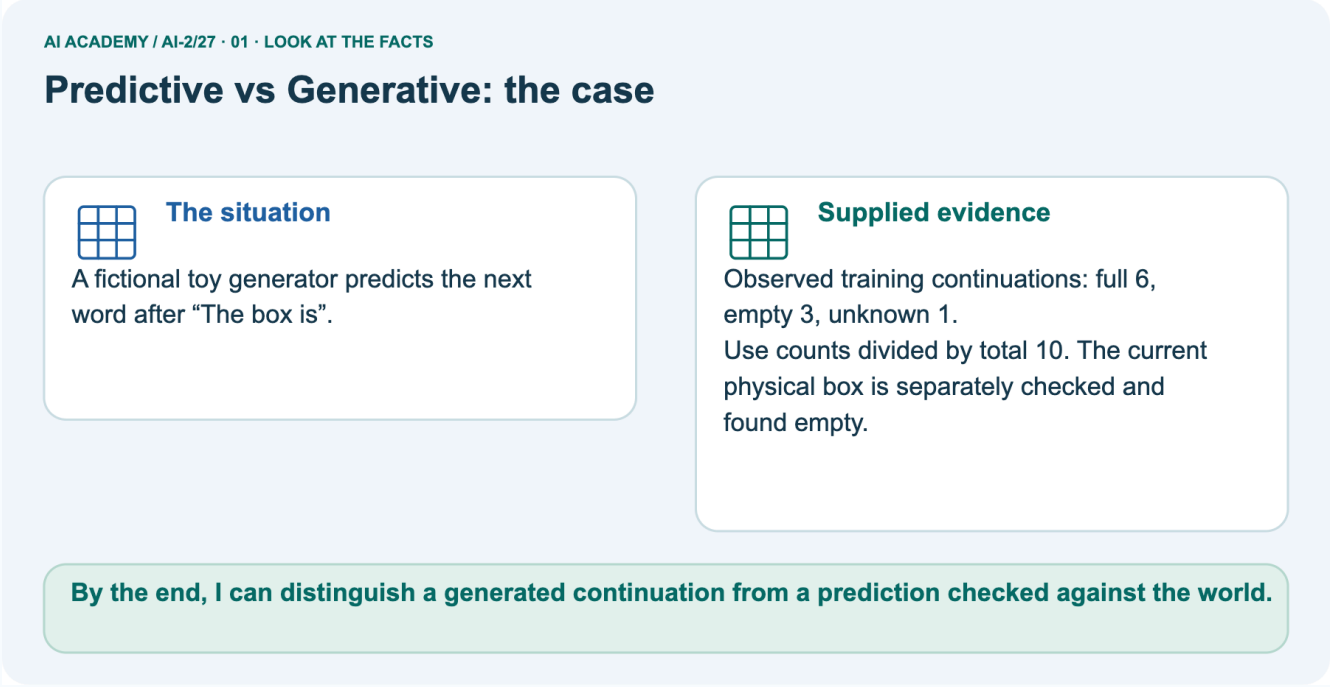
▼ **Reference: the six-picture journey**

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Predictive vs Generative: the case



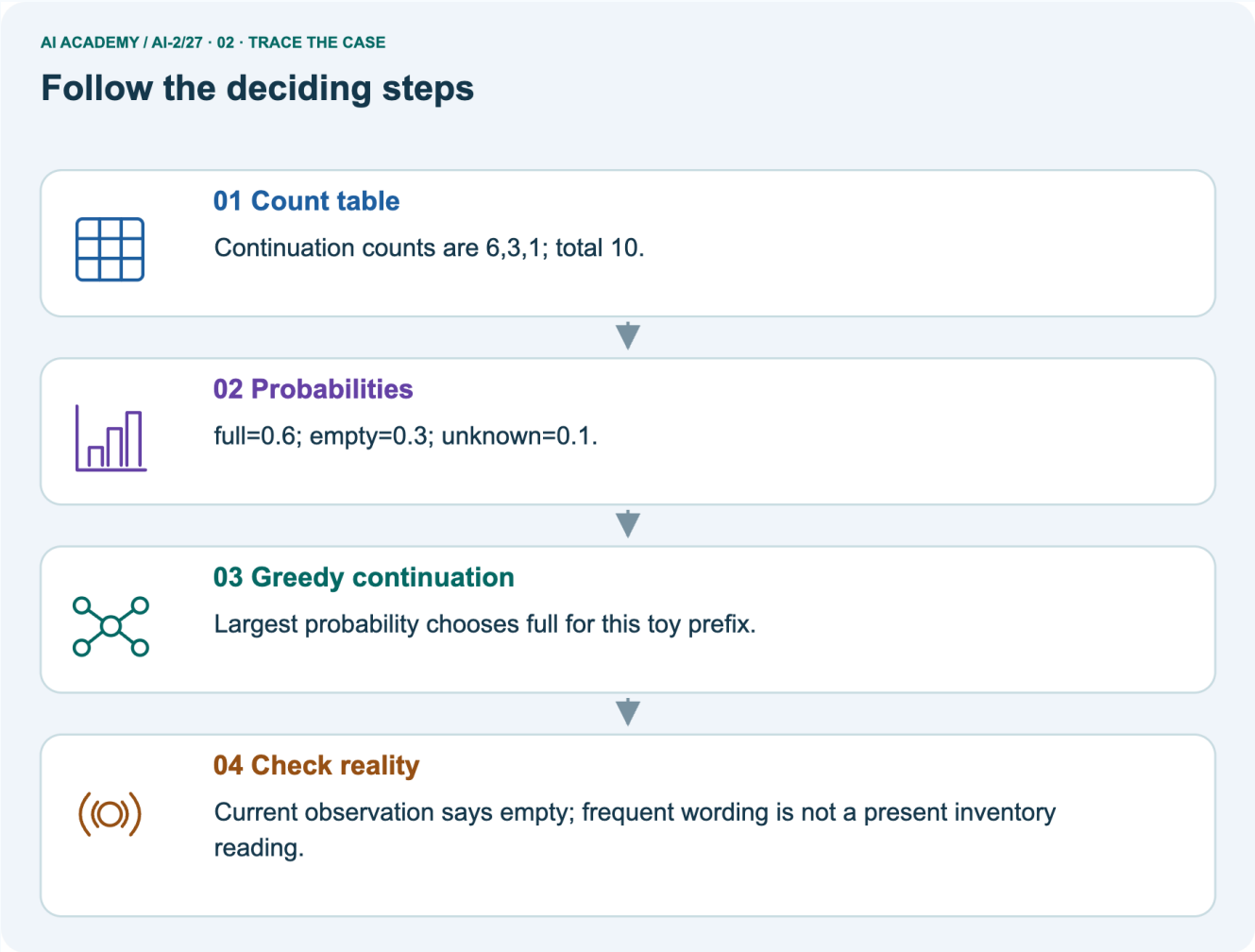
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional toy generator predicts the next word after “The box is”. Supplied facts: Observed **training** continuations: full 6, empty 3, unknown 1. Use counts divided by total 10.

The current physical box is separately checked and found empty.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Count table: Continuation counts are 6,3,1; total 10. → Probabilities: full=0.6; empty=0.3; unknown=0.1. → Greedy continuation: Largest probability chooses full for this toy prefix. → Check reality: Current **observation** says empty; frequent wording is not a present inventory reading.

Point to each numbered stage and explain which recorded **input** it uses.

03. Toy next-word distribution

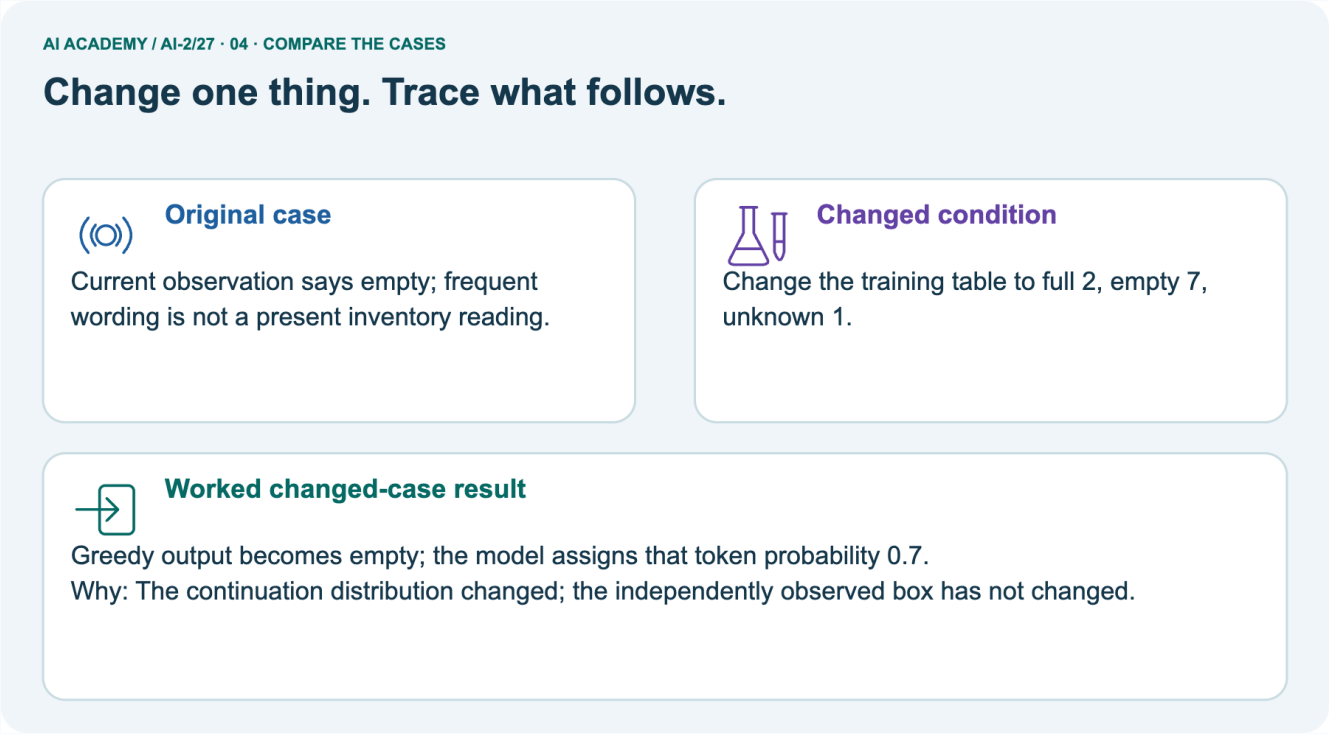


Follow the labelled picture with the matching explanation. Open the vector diagram

Toy next-word distribution. Units: probability. full: 0.6. empty: 0.3. unknown: 0.1.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



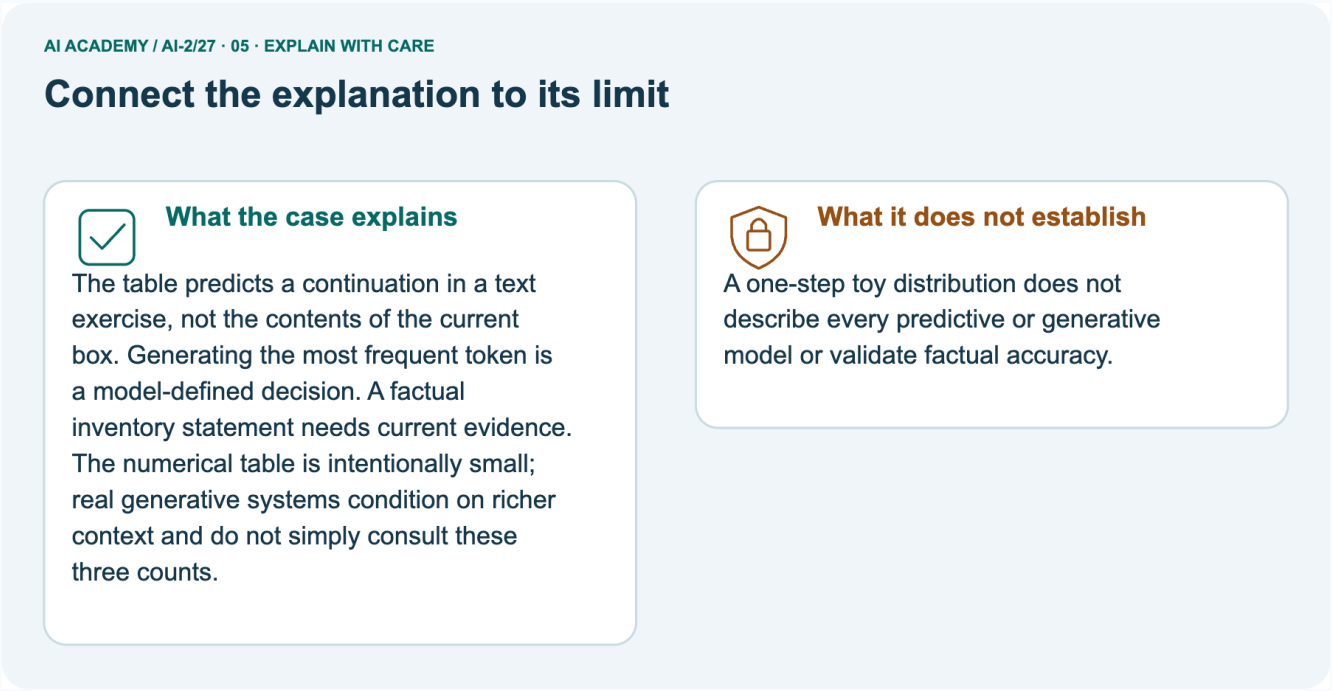
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Current **observation** says empty; frequent wording is not a present inventory reading. Changed condition: Change the **training** table to full 2, empty 7, unknown 1.

Expected result: Greedy **output** becomes empty; the model assigns that token probability 0.7. Reason: The continuation distribution changed; the independently observed box has not changed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

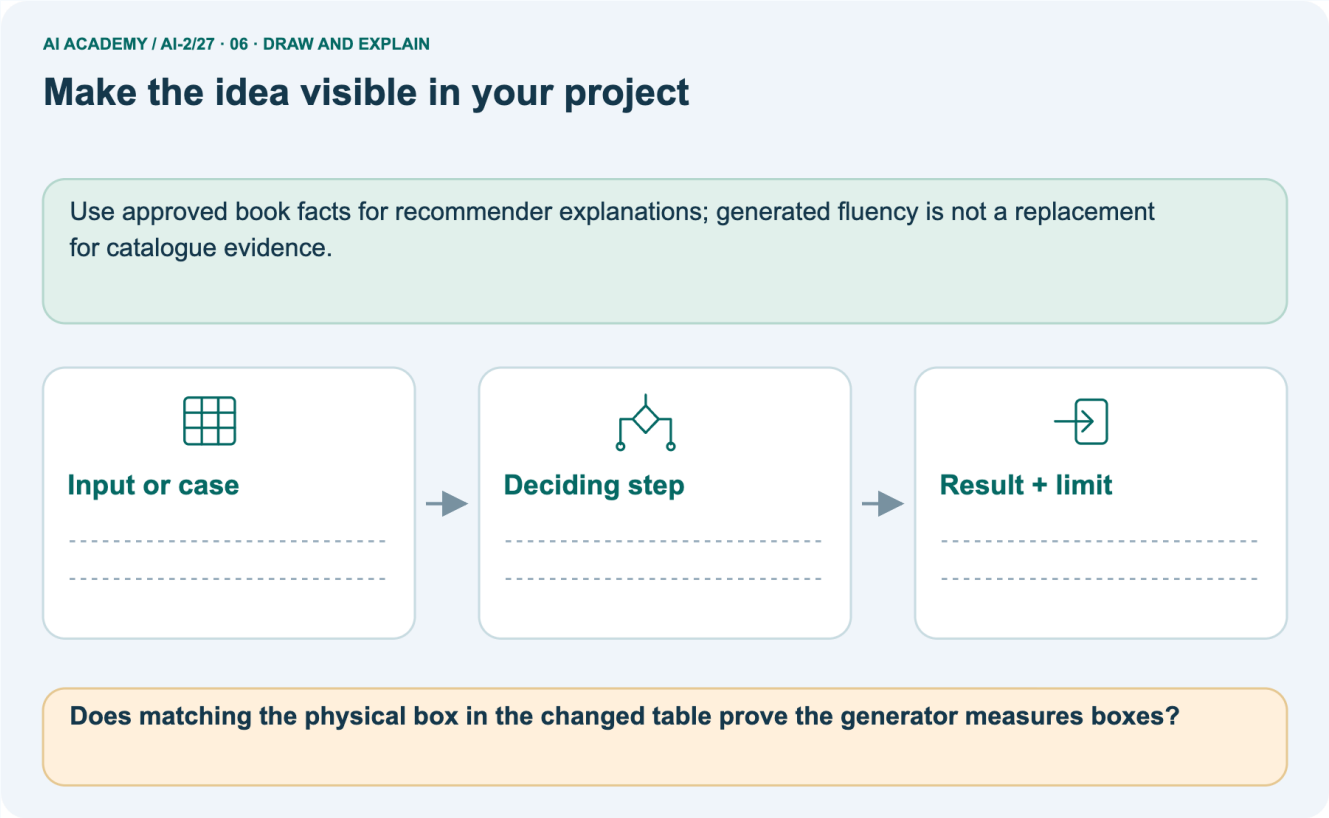
The table predicts a continuation in a text exercise, not the contents of the current box. Generating the most frequent token is a model-defined decision. A factual inventory statement needs current **evidence**.

The numerical table is intentionally small; real **generative** systems condition on richer context and do not simply consult these three counts.

Limit: A one-step toy distribution does not describe every **predictive** or **generative** model or validate factual accuracy.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use approved book facts for recommender explanations; generated fluency is not a replacement for catalogue **evidence**. Draw your **input**, deciding step, result and limit.

Supported question: Does matching the physical box in the changed table prove the generator measures boxes?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Predictive vs Generative: the case

02. Follow the deciding steps

03. Toy next-word distribution

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.